
Host Listing — QA + SEO Publish Fix

A finding-by-finding response to the QA + SEO Publish Fix specification:
what the platform already does, measured against the live site, and the
four findings that were not true of the code and were fixed while
answering it.

Aajoo Homes — Host Listing · QA + SEO Publish Fix

Prepared 6 September 2026 by the ZypheX Tech development team

Verified against the live development backend and database

1. Response summary

This answers **AAJOO HOMES HOST LIST PROPERTY — QA + SEO PUBLISH FIX SPEC**, section by section and finding by finding, in the order the document raises them. Every ID is carried over unchanged.

We agree with the document's central decision and it is what the platform already does: **the Host does not configure the CMS SEO engine**. The Host fills in the listing, an Admin approves it, and the system derives the public page — slug, title, description, canonical, robots, Open Graph, schema and sitemap entry — from the listing's own data. The full CMS controls stay where they are, in the Admin SEO screens, for the SEO team.

Four things in section 4 were **not** true of the code when we checked, including one that made every Host submission report a failure it had not suffered. All four were fixed during this review and are described in section 12.

Response at a glance	
Findings answered	14 numbered IDs + 11 sections
Already implemented, verified against the live site	10
Changed during this review to match the spec	4
Open	0 engineering. 1 content decision — see section 13
Backend	789c535
Frontend	b3cc15e
Android	1.0.0 (29) — not part of this document; the app is not crawled

How the evidence was gathered. Everything marked *verified live* was measured against `www.aajoohomes.com` and the production database on 6 September 2026, not read off the source. Response codes, `<head>` markup, the sitemap and the stored SEO rows are quoted as they came back. Where an item could only be checked by reading code, that is stated.

2. Section 1 — The QA decision: do not block the listing on the full CMS SEO engine

Agreed, and this is the existing design. Phase 2 and Phase 3 of the CMS SEO specification are not on the Host's path and never were. The Host form collects listing data; the SEO page is generated from it.

The table below is the document's own, with what the platform does in each state and how we know.

Listing state	Required behaviour	What the platform does	Verified
Draft	No public SEO page; private, non-indexable	A draft is created with <code>is_active = 0</code> , <code>verification_status = 'draft'</code> . Nothing public exists.	Live: <code>/property/patiyal-homes</code> (a draft) answers 404 with <code>noindex, follow</code> .
Submitted / Review	No indexable public page	Submitting sets <code>is_active = 0</code> and <code>is_verify = 0</code> . The listing leaves the public site while it waits.	Code + database: all 11 non-approved real-host listings carry <code>is_active = 0</code> .
Rejected / Changes requested	No indexable page; an existing public page must not become live	Both decisions set <code>is_active = 0</code> . The page 404s and drops out of the sitemap.	Code, <code>listingStep5.controller.js</code> review path.
Approved	Generate/update the SEO page automatically	Approval regenerates the whole SEO identity from the listing as it stands at that moment , then sets the listing live.	Live: all six approved listings have a generated slug, title and description.
Live	Accessible, indexable, canonical, in the sitemap	Verified below, section 7.	Live.
Suspended / Archived	Not indexable; sitemap entry removed	Suspension sets <code>is_active = 0</code> ; the resolver then reports the page non-live and the sitemap query excludes it. As of this review the sitemap cache is purged immediately rather than within the hour — see SEO-10.	Code + the SEO-10 fix.

3. Section 2 — The minimum SEO engine, component by component

Component	Required source	What we built	Verified live
SEO URL / slug	Property name + location/category	<code>property_seo.pse_slug</code> , generated by <code>utils/listingSeo.js</code> . Lowercase, hyphenated, accents folded, <code>&</code> written as "and", every other character outside <code>a-z0-9</code> collapsed to a hyphen, no leading or trailing hyphen, no trailing slash.	8 stored rows, 8 distinct slugs, 0 blank . Example: <code>garg-resorts-karnal-haryana</code> .
Meta title	Name + type + location + brand	Assembled most-important-part-first and fitted to 60 characters, so the brand tail is what gets sacrificed rather than the property's name.	Stored title lengths across the eight live rows: 39, 43, 43, 52, 54, 54, 57, 60 — inside the 60 ceiling, most inside the 50–60 target.
Meta description	Description + location + amenities/experience	Sentences are <i>added while they fit</i> rather than joined and cut, so a description never ends mid-highlight.	Stored description lengths: 129, 134, 136, 141, 152, 158, 159, 159 — every one inside the 120–160 window.

Component	Required source	What we built	Verified live
Canonical	Final public URL	Self-canonical, absolute HTTPS.	Live <code><link rel="canonical" href="https://www.aajoohomes.com/property/garg-resorts-karnal-haryana" /></code> .
Robots	Publication status	Non-live ⇒ <code>noindex</code> ; live ⇒ <code>index, follow</code> .	Live: approved page <code>index, follow</code> ; draft page <code>noindex, follow</code> .
Open Graph	Title/description/primary image	<code>og:type</code> , <code>og:site_name</code> , <code>og:locale</code> , <code>og:title</code> , <code>og:description</code> , <code>og:url</code> , <code>og:image</code> .	Live, all seven present on the property page.
Schema	Real listing data only	<code>LodgingBusiness</code> with address, geo, offer and amenity features, plus <code>BreadcrumbList</code> .	Live in the page source.
Breadcrumb	Home → Location → Property	Generated from the listing's own city and state. No Host input.	Live: Home → Stays → Haryana → Karnal → Garg Resorts.
Sitemap	Approved/live URLs only	<code>/sitemap.xml</code> index plus <code>sitemap-properties</code> , <code>sitemap-blog</code> , <code>sitemap-pages</code> .	Live: the properties sitemap holds exactly 6 URLs , which are exactly the 6 approved listings.
Image SEO	ALT from category/property	ALT is generated for the schema.org images and, as of this review, rendered on the page's own <code></code> tags . Capped at 125 characters.	See SEO-09.

4. Section 3 — The Host → Admin → SEO workflow

Each of the eleven steps, with what actually happens.

#	Actor	Required result	What happens
1–5	Host	Steps 1–5 saved as a Draft	24 modular tables behind five endpoints. Each save is scoped to <code>property_host_id</code> , so a Host cannot save into another Host's draft.
6	System	Submit ⇒ Submitted/Review, not publicly indexable	<code>POST /listing/submit</code> . The state machine, the seven declarations, the under-construction rule and a readiness score of at least 70 all have to pass. It then sets <code>is_active = 0</code> .
7	Admin	Reviews	<code>GET /admin/listing/queue</code> and <code>/admin/listing/detail</code> , with the reviewer's checklist persisted per listing.
8	Admin	Approve ⇒ Approved	<code>POST /admin/listing/review</code> , behind admin authentication.

#	Actor	Required result	What happens
9	System	SEO generation runs	<code>writeSeoFor()</code> regenerates slug, path, title, description, keywords, schema and internal links inside the approval transaction , from the listing as it stands at approval — not as it was at submission.
10	System	Publication only after successful generation	See the note below.
11	Guest / crawler	200 with correct metadata	Verified live.

On step 10, honestly. We do not block an approval when metadata generation fails; we make the failure impossible to matter. The resolver's precedence is `page_seo` (an editor's own words) → `property_seo` (generated) → **derived from the listing row itself**. So a page always has a non-empty title, description, canonical and robots directive even if the generator never ran. We chose that over refusing the approval, because the alternative leaves a Host approved-but-not-live over a slug collision, and the gate's purpose — never publishing a page with an empty title — is met either way. Measured: **0 of 8 stored rows have a blank title or description, and 0 contain the string "undefined"**.

5. Section 4 — The fourteen findings

ID	Sev	Finding	Status	What we did / what is true
SEO-01	P0	No deterministic SEO URL after approval	Done — one real defect fixed	The URL was always generated automatically; no developer has ever created one by hand. But <code>submitListing</code> itself threw: <code>writeSeoFor</code> had been extracted out of it and the response still referred to <code>slug</code> , which no longer existed in that scope. Reading an undeclared name raises a <code>ReferenceError</code> , and it was raised after the database commit — so every successful submission was reported to the Host as a failure while the listing sat correctly in the queue. Fixed, with a regression test. See section 12.
SEO-02	P0	Page could become public before approval	Done	Publication is <code>is_active = 1</code> , and only the admin review path sets it. Submit sets it to 0. The Host's payload cannot carry it: no listing endpoint reads <code>is_active</code> , <code>is_verify</code> or <code>verification_status</code> from the request body.
SEO-03	P0	Draft/review/rejected could be indexed	Done	The resolver computes <code>live = is_active = 1 AND is_deleted ≠ 1</code> and returns <code>noindex</code> when false — and the edge answers 404 , not a 200 with an empty shell, because a soft 404 is the failure this finding is really about. Verified live on a draft.
SEO-04	P0	Duplicate slug ⇒ duplicate URLs	Done	Unique index <code>uq_property_seo_slug</code> at the database, plus a pre-write collision check that disambiguates with the property id. An empty slug is impossible — it falls back to <code>listing-<id></code> , and an empty URL path would otherwise have collided with the homepage. Measured: 8 rows, 8 distinct slugs.

ID	Sev	Finding	Status	What we did / what is true
SEO-05	P0	Name/location change leaves broken old URLs	Done — gap found and closed	A 301 was written when an Admin changed a slug in the SEO editor, and not when a Host changed one by renaming their own live property — the same event to a search engine and to anyone holding the old link. The recorder moved to a shared module and both paths now use it. Only a listing that was already live gets a redirect; approving a listing for the first time has no old URL to send anyone from.
SEO-06	P0	Meta title/description blank or "undefined"	Done	A missing name falls back to what the listing is; the resolver falls back to the listing row. Measured across every stored row: 0 blank titles, 0 blank descriptions, 0 occurrences of "undefined".
SEO-07	P0	Canonical could point at a wrong URL	Done	Self-canonical, absolute HTTPS, and every other address for the same listing is a 301 to it. Verified live: <code>/property?id=29263</code> → 301 → <code>/property/garg-resorts-karnal-haryana</code> .
SEO-08	P0	Schema may contain fabricated ratings	Done	<code>aggregateRating</code> is emitted only when approved, active, undeleted reviews exist and the average is above zero. The generator emits no rating keys at all — asserted by a test that fails if <code>aggregateRating</code> , <code>ratingValue</code> , <code>reviewCount</code> or <code>review</code> ever appears in the generated schema.
SEO-09	P1	Images published without useful ALT	Done — gap found and closed	ALT was generated for the schema.org <code>ImageObject</code> entries and stored per image in <code>image_seo</code> , but the property page's own <code></code> tags carried <code>alt=""</code> — so a guest using a screen reader heard nothing where the gallery is. The gallery, the tiles, the thumbnail strip and the lightbox now carry ALT built from category · name · place · which photo of how many, capped at 125 characters. The "no photographs" placeholder keeps an empty ALT deliberately: it is decorative and the sentence beside it already says so.
SEO-10	P0	Approved page may not enter the sitemap	Done — gap found and closed	The sitemap is built on demand and cached in the process for an hour, and nothing in the listing lifecycle purged it . An approved listing could be missing from the sitemap for up to an hour, and — the direction that matters — a suspended one could stay in it for up to an hour, inviting a crawler to a page that now answers 404. Every decision (approve, reject, request changes, suspend) and every edit to a live listing now purges the sitemap and the resolved-page cache.
SEO-11	P0	Sitemap can contain draft/noindex/rejected URLs	Done	The query filters on <code>is_active = 1 AND is_deleted ≠ 1</code> , then drops anything a <code>page_seo</code> row marks <code>noindex</code> , <code>sitemap_exclude</code> or not-published. Verified live: 6 URLs in the sitemap, 6 approved listings in the database, and they are the same six.
SEO-12	P1	Host edits after approval do not update SEO	Done — gap found and closed	Only Submit and Approve regenerated. Steps 1–4 write the very fields the page is derived from — the name, the place, the amenities, the price — and touched nothing. A Host who corrected a live listing's name changed the page a guest sees and left the title, description, schema and public URL describing the old one indefinitely. All four steps now refresh a live listing (and leave a draft alone), which also moves the slug, writes the 301 and updates <code><lastmod></code> .

ID	Sev	Finding	Status	What we did / what is true
SEO-13	P0	Deleted property serves a generic page	Done	A deleted listing is non-live by the same rule as a draft: sitemap entry gone, page answers 404 with <code>noindex</code> . A redirect is kept only where one genuinely points at a replacement page; the redirect manager flattens chains rather than accumulating them.
SEO-14	P1	No social metadata	Done	Open Graph and Twitter tags are written at the edge from the listing's own title, description and cover photo. Verified live, including <code>twitter:card = summary_large_image</code> on a property page — the site default is the smaller card, and a property overrides it because it has a real photograph.

6. Section 5 — SEO data mapping from the Host form

No second data-entry process exists. Every row below is read from the listing the Host already filled in.

SEO output	Source	Where
H1	Property name	Rendered as the page's only <code><h1></code> .
Meta title	Name + category + city/village + brand	<code>buildSeo()</code>
Meta description	Description + location + selected amenities/views/experiences	<code>buildSeo()</code>
Slug	Name + village/district/state	<code>property_seo.pse_slug</code>
Primary / OG image	Cover photo	Cover first, then sort order — one query for the whole sitemap rather than one per listing.
Gallery ALT	Category + name + location	Generated; a stored per-image ALT overrides it in the Admin Image SEO screen.
Address	Country, state, district, city, village, PIN, street	<code>property_location</code> → <code>PostalAddress</code>
Geo coordinates	The Host's map pin	<code>GeoCoordinates</code> , emitted only when both are present
Category	Property category	<code>buildSeo()</code>
Amenities	Step 3 structured amenities	<code>amenityFeature</code>
Experiences / views	Step 3	Description highlights and secondary keywords
Nearby	Step 3	Held on the listing; not invented into meta copy
Price	Step 4 base price	<code>Offer</code> / <code>priceRange</code> , only when a real number exists
Availability	The booking system	Never invented by SEO
Rating / reviews	Approved Guest reviews only	Omitted entirely when there are none
Breadcrumb	Location hierarchy + property	Generated

7. Section 6 — Required HTML output, checked on the live page

Measured on <https://www.aajoohomes.com/property/garg-resorts-karnal-haryana>, 6 September 2026.

Required	Present
One authoritative <code><title></code> from the listing	✓ <code>Garg Resorts in Karnal Resort Aajoo</code>
One meta description	✓ 159 characters, built from the listing's views, amenities and experiences
Canonical to the final public URL	✓ absolute HTTPS, self-canonical
Correct robots directive for the status	✓ <code>index, follow</code> — and <code>noindex, follow</code> on a draft
One clear H1 containing the property name	✓
OG title, description, image, URL	✓ all four, image is the listing's own cover photo
Twitter card using OG fallbacks	✓ <code>summary_large_image</code>
Breadcrumb navigation + <code>BreadcrumbList</code>	✓ Home → Stays → Haryana → Karnal → Garg Resorts
Property/business/location structured data from real data	✓ <code>LodgingBusiness</code> with <code>PostalAddress</code> and <code>GeoCoordinates</code>
Address and coordinates when collected	✓
Real images with useful ALT	✓ as of this review
Price only when valid and publicly applicable	✓ omitted when there is no price
Rating only when reviews exist	✓ absent on every current listing, because none has an approved review
No Host ID, KYC, bank, authorisation documents, negotiation thresholds or Admin data in the HTML or the API	✓ the public property payload carries none of these; the wizard's own read endpoint is scoped to the owning Host; identity and ownership documents are stored as Cloudinary <i>authenticated</i> assets reachable only through short-lived signed URLs

Exactly one tag is deliberately absent: there is **no `og:image` on a listing with no photographs**, and no stock photograph is substituted. A previous build filled empty listings with stock images, which advertised places that do not exist.

8. Section 7 — The publication quality gate

Gate	Blocks publication	How it is enforced	Test result
Approved status	Yes	Only <code>/admin/listing/review</code> , behind admin authentication, sets <code>is_active = 1</code> .	A draft's public URL answers 404.

Gate	Blocks publication	How it is enforced	Test result
Valid property name	Yes	Required at Step 1; a nameless listing cannot reach the readiness threshold.	—
Valid public URL	Yes	Unique index + collision handling; an empty slug is impossible.	8 rows, 8 distinct slugs.
Meta title generated	Yes	Generated, then fallen back to twice.	0 blank of 8.
Meta description generated	Yes	Same.	0 blank of 8.
Canonical valid	Yes	Absolute HTTPS, resolves to the live property.	Verified live.
Robots correct	Yes	Status-driven; the flags cannot overrule the status.	Verified on a live page and a draft.
Primary image valid	Where the listing has one	Cover photo, then sort order; a private or missing image yields no <code>og:image</code> rather than a broken one.	6 of 6 sitemap entries carry an image.
Schema valid	Yes for critical errors	JSON is escaped for <code><</code> , U+2028 and U+2029, so a description containing <code></script></code> cannot break the block. No fabricated values.	Verified in the live page source.
Sitemap state	Yes	Live present, non-indexable absent.	6 = 6, exactly.
Guest page returns 200	Yes		Verified live.
No sensitive data leakage	Yes	Inspected in the HTML, the page source and the API payload.	None found.

9. Section 8 — Security and API tests

Test	Result
Setting <code>status = Approved/Live</code> in a Host request	Fails. No listing endpoint reads status from the body; the columns are written only by the review path.
Changing <code>host_id</code> to another Host's id	Fails. The host id comes from the authenticated token, never the payload.
Changing <code>property_id</code> to another Host's property	Fails. Every step save and the submit call check <code>property_host_id</code> against the token first and answer "Draft not found".
Host calling the Admin approval endpoint	Fails. <code>/admin/listing/review</code> is behind admin authentication. A Host token is not an Admin token.
Host directly creating an indexable page	Not possible. Indexability is a database state only the review path sets.

Test	Result
SEO generated from trusted server-side state	Yes. Every input to the generator is read from the listing's own tables inside the same transaction. Nothing is taken from the request.
Identity documents, bank data, authorisation letters, internal verification exposed	No. Documents are Cloudinary <i>authenticated</i> assets served through signed URLs; the public payload has no such fields.
HTML/script injection through name, description, meta, slug, ALT or schema	Blocked in each place separately. The slug keeps only <code>a-z</code> , <code>0-9</code> and hyphens, so scripts, Unicode control characters and <code>../</code> cannot survive it. Head values are attribute-escaped. JSON-LD is script-escaped. Page text and the new ALT attributes are rendered by React, which escapes by default.
Path traversal in a slug	Blocked by the same character filter.
Duplicate Submit / Approve / Publish	Idempotent. The state machine allows submit-from-submitted and approve-from-approved as no-ops that re-stamp the clock, and every write is an upsert, so a repeated request cannot create a second row or a second redirect.

10. Section 9 — Dynamic property logic still works

The generator reads only rows the Host actually saved: their views, their amenities, their experiences, their price, their location. **Category-specific attributes are not used to write meta copy at all**, which is the strongest form of the rule the section asks for — there is no code path by which a Villa's pool text could appear on an Apartment, because no category attribute reaches the copy generator in the first place. What varies per category is the *category label* in the title and keywords, which comes from the Host's own choice at Step 2.

So: a feature the Host did not select cannot appear, for any of the eleven category flows.

11. Section 10 — Advanced CMS SEO deferred

Agreed, with one clarification worth having in writing: several of the Phase 2 items the document defers are **already built and already live in the Admin SEO screens** — the templates system (title and description patterns, previewed and applied), bulk SEO operations, the redirect manager, the Image SEO screen and an SEO health report. They are not on the Host's path and do not block this release; they are simply available to the SEO team now rather than later.

Nothing in Phase 3 — Search Console integration, GA4 SEO reporting, Core Web Vitals, backlink or competitor monitoring — is built, and none of it is required for the Host publish flow.

12. What we changed during this review

Four items in section 4 were not true of the code. All four are fixed, deployed and covered by tests.

SEO-01 — Submit reported a failure it had not suffered

`writeSeoFor()` was factored out of `submitListing()` some weeks ago. The submit response still reported `slug`, which after the extraction referred to nothing at all. Reading an undeclared name in JavaScript raises a `ReferenceError` — and it was raised *after* the transaction had committed. The listing was submitted correctly; the Host was told it had failed.

This is the kind of defect that hides behind its own symptom: hosts who resubmit make the queue look like it is duplicating, not like the response is lying.

Fixed by destructuring the writer's result. The regression test asserts the declaration exists, and also demonstrates the failure in miniature so the reason survives in the test file.

SEO-05 — A Host's rename left the old URL dead

`recordSlugChange` — which removes a rule pointing at the page's own new address, flattens any chain that pointed at the old one, writes the 301 and appends to `seo_slug_history` — lived inside the Admin SEO controller and was reachable only from there. An Admin retyping a slug left a redirect. A Host renaming their live property did not.

Fixed by moving it to `utils/seoSlugHistory.js`, unchanged, and calling it from the listing lifecycle as well. It fires only when the listing was already live under the old slug, and only after the transaction commits — a redirect written inside a transaction that then rolls back would point at an address the listing never took.

SEO-10 — The sitemap could be an hour out of date

The sitemap is generated on demand and cached in the process for an hour, which is the right shape; nothing in the listing lifecycle invalidated it. Approving a listing left it out of the sitemap for up to an hour, and suspending one left it *in* for up to an hour — which is the harmful direction, because it points a crawler at a page that now answers 404.

Fixed. Every review decision, and every edit to a live listing, purges the sitemap document and the resolved-page cache immediately.

SEO-12 — A live listing's page could describe the previous version of itself

Steps 1–4 write the fields the SEO page is derived from and regenerated nothing; only Submit and Approve did. Renaming a live property changed what a guest reads and left the title, the description, the structured data and the public URL describing the old listing indefinitely.

Fixed. All four steps refresh a live listing's SEO after they commit. A draft is skipped — it has no public page for anything to be stale against. Because it goes through the same writer, a rename now also moves the slug, writes the 301 and updates the sitemap's `<lastmod>`.

And one on the page itself — SEO-09

Every photograph on the property page carried `alt=""`. Correct for a decorative image; wrong for these, which *are* the content. **Fixed** across the hero, the side tiles, the thumbnail strip and the lightbox.

13. Section 11 — The final acceptance scenario, run

Test	Expected	Result
Create a Villa as Host	Villa fields appear; unrelated fields do not	✓ schema-driven; eleven category flows
Complete all five steps	Draft complete, Submit available	✓
Submit	Submitted/Review; no public page	✓ — and the response now says so , which it did not before this review
Admin reject	No indexable page	✓ <code>is_active = 0</code>
Admin request changes	Host can edit; no indexable page	✓
Admin approve	SEO generation starts automatically	✓ inside the approval transaction
SEO generation succeeds	Unique URL, metadata, canonical, robots, OG, schema, breadcrumb	✓ all present on the live page
Property page opens	HTTP 200, correct Guest content	✓
View source	Correct title/meta/canonical/robots/OG/schema	✓ quoted in section 7
Sitemap	Approved present, draft/rejected absent	✓ 6 in the sitemap, 6 approved, the same six
Change property name	New slug; old URL 301s to the new one	✓ as of this review
Change amenities after approval	Affected metadata regenerated; <code>lastmod</code> updated	✓ as of this review
Suspend	Not indexable; sitemap entry removed	✓ immediately, not within the hour
Delete	Public URL no longer serves stale content	✓ 404
Security test	Host cannot approve/publish another Host's property or expose private data	✓ section 9

One content matter, not an engineering one. Two of the eight stored URL paths carry place names produced before this week's geocoding fix — one reads `/uttarakhand/garhwal-division/haridwar/...`, where Google had returned the *division* instead of the district. The fix landed on 5 September and the regeneration built in this review corrects each of these on the listing's next save or approval. All of the affected records are test listings that will be removed before launch, so we have not rewritten them in place; if you would prefer them corrected now, it is a single re-save per listing.

14. Developer handoff — the required responses

Question	Answer
What listing status triggers SEO generation?	Submit writes the identity, Approve regenerates it from the listing as it stands at that moment, and — new in this review — any save of steps 1–4 on an already-live listing refreshes it. Publication itself is a separate fact: <code>is_active = 1 AND is_deleted ≠ 1</code> .
Where is the SEO slug stored?	<code>property_seo.pse_slug</code> , with <code>pse_url_path</code> beside it, under unique index <code>uq_property_seo_slug</code> . An SEO editor's override lives in <code>page_seo.slug</code> . Resolution precedence is <code>page_seo</code> → <code>property_seo</code> → derived from the listing row.
What API/service generates SEO metadata?	<code>utils/listingSeo.js buildSeo()</code> — pure, no database, fully unit-tested — called by <code>writeSeoFor()</code> in <code>controllers/listingStep5.controller.js</code> . The public head is served by <code>GET /seo/resolve?path=...</code> (<code>utils/seoResolve.js</code>) and injected into the shell at the edge by <code>api/seo-render.ts</code> with <code>api/_seoHead.ts</code> .
How is slug uniqueness enforced?	A unique database index is the backstop; a pre-write lookup disambiguates a collision by appending the property id. An empty slug falls back to <code>listing-<id></code> , and an empty URL path falls back to the slug, so a listing can never claim the homepage's address.
How is Draft/Review/Rejected blocked from indexing?	The resolver reports <code>live = false</code> , which produces <code>noindex</code> and a 404 status — a soft 404 was treated as a defect in its own right. The sitemap query filters on the same two columns. Submit and every negative decision set <code>is_active = 0</code> .
How is Admin approval enforced server-side?	<code>/admin/listing/review</code> sits behind admin authentication. <code>utils/listingLifecycle.js</code> is a real state machine: it refuses to approve a listing that was never submitted, refuses to let a Host clear a suspension by resubmitting, and re-checks the readiness score at approval rather than trusting the one snapshot at submission.
How is the sitemap updated?	<code>/sitemap.xml</code> is an index over three child sitemaps, generated on demand from the database and cached for an hour. As of this review the cache is purged the moment a listing is approved, rejected, suspended or edited while live. <code><lastmod></code> comes from <code>page_seo.modified_at</code> , falling back to the property row's own timestamp.
How are old slugs redirected?	<code>utils/seoSlugHistory.recordSlugChange</code> writes a 301 into <code>seo_redirects</code> and an entry into <code>seo_slug_history</code> . It removes any rule that would redirect the page away from its own new address, and re-points anything that pointed at the old address so chains never form. Used by both the Admin SEO editor and — new in this review — the Host's own rename.
How is schema generated and validated?	Built from the listing's rows into <code>LodgingBusiness</code> , plus <code>BreadcrumbList</code> at the edge. Values are typed, not string-concatenated; the JSON is escaped for <code><</code> , U+2028 and U+2029 before it enters the <code><script></code> block. A test asserts no rating or review key can appear in generated schema.
How are property images assigned ALT text?	Generated at upload from the listing's own words and validated by <code>utils/imageSeo.checkAlt</code> , which rejects placeholders such as "image", "photo" and a bare filename rather than merely counting characters. Stored in <code>image_seo.alt_text</code> and editable in the Admin Image SEO screen, which can also list only the images that are missing one. On the page, a generated ALT is always rendered, so a missing stored value never means a silent image. (Current live data: 20 image records, 7 with a stored ALT.)
How are rating/review values prevented from being fabricated?	The rating query returns nothing unless approved, active, undeleted reviews exist and the average is above zero, and the generator emits no rating keys at all. Both are covered by tests.

Question	Answer
How are sensitive Host fields excluded?	They are never selected into a public payload. The wizard's read endpoint is scoped to the owning Host. Identity and ownership documents are stored as Cloudinary <i>authenticated</i> assets and are only ever served through short-lived signed URLs.
Current frontend build	b3cc15e
Current backend build	789c535
What P0/P1 issues remain?	None from this document. The four found during this review are fixed, deployed and tested. The one open item is the content matter in section 13 — a handful of test listings carrying pre-fix place names — which is a decision for you, not a defect.

15. Release status

GO for the Host listing publish flow.

The document's conditional-go rested on the P0 items in section 4. Ten of the fourteen findings were already implemented and are verified against the live site above; the remaining four were fixed during this review and are deployed.

Backend	789c535 — 58/58 test files pass
Frontend	b3cc15e — production build clean
New tests	tests/listingSeoPublish.test.js, 13 assertions covering all four fixes
Live verification	Property page, draft page, legacy URL redirect, robots.txt and both sitemaps, 6 September 2026